

Role of N-terminal domain of histidine-rich glycoprotein in modulation of macrophage Fc γ receptor-mediated phagocytosis

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SUMMARY

A brief exposure of murine peritoneal inflammatory macrophages to plasma histidine-rich glycoprotein (HRG; 77,000–81,000 MW) for 1–2 hr increased Fc γ receptor (Fc γ R) expression and phagocytic function in these cells. However, a continual culture of the cells without the presence of HRG for the next 18–48 hr resulted in down-regulation of Fc γ R expression and phagocytic function. Similarly, HRG decreased Fc γ RII expression in less differentiated human THP-1 monocytic cells during treatment for 18 hr, as determined by cellular ELISA and metabolic labelling. The molecular mechanism by which HRG regulates Fc γ R expression is unknown. However, at a relatively high concentration (>1 μ g/ml), HRG altered the cellular metabolism by increasing cellular protein synthesis but reducing protein secretion. These observations suggest a likely mechanism for the HRG-mediated reduction of Fc γ R expression. A degraded HRG (40,000 MW) which possessed an identical N-terminal sequence as that of the native HRG was capable of decreasing macrophage Fc γ R expression and phagocytosis. The results indicate that the functional domain of HRG responsible for binding to macrophages is localized to the N-terminal half.

INTRODUCTION

Histidine-rich glycoprotein (HRG) is a single-chain polypeptide (77,000–81,000 MW; 507 amino acid residues) with a substantial concentration (100–150 μ g/ml) in plasma. Biochemically, HRG is characterized by an unusually high content of proline and histidine, which comprises higher than 12% of the total amino acids of the protein.^{1–3} The primary structure of HRG deduced from the cDNA is shown to possess 12-tandem repetitions of a 5-amino acid segment with a consensus sequence of Gly-His-His-Pro-His (residue no. 330–389), which is referred to as the histidine-rich region.¹ The *in vivo* function of HRG has not been well defined. However, due in part to its highly charged histidine region, HRG effectively binds divalent cations,⁴ heparin^{5–7} and heme.⁸ Additionally, HRG forms complexes with platelet thrombospondin,⁹ plasma plasminogen² and S-protein (vitronectin).¹⁰ HRG regulates the functional efficiency of the coagulation/fibrinolysis system by interactions with heparin, plasminogen and probably other components. HRG also could modulate serum complement activation of both the classical and alternative pathways, in which it has been shown to interact with complement C8, C9 and D and S-protein.¹⁰

In vitro studies have shown that HRG inhibits auto-rosette

formation between T lymphocytes and autologous erythrocytes,^{11–13} and restricts proliferation and interferon- γ (IFN- γ) production in T cells.¹⁴ Furthermore, HRG modulates murine peritoneal inflammatory macrophage Fc γ receptor (Fc γ R) expression and phagocytosis of antibody-opsonized sheep erythrocytes (EA).¹⁵ HRG up-regulated macrophage Fc γ R expression and phagocytosis during an initial pretreatment for 1–3 hr, followed by down-regulation when the pretreatment was prolonged for up to 16–24 hr. In the present study, we have further characterized HRG modulation of macrophage functions and localized its functional domain to the N-terminal half. Modulation of cellular protein synthesis and/or secretion by HRG appears to be correlated with changes of macrophage Fc γ R expression and phagocytic function.

MATERIALS AND METHODS

HRG purification and gas-phase micro-sequencing

Human or mouse plasma HRG was purified according to an established procedure, as described previously.¹⁵ The chromatographic purification steps included concanavalin A (Con A)–Sephacrose 4B, heparin–agarose and DEAE–Sephacel columns. The purified protein of 77,000–81,000 MW was subjected to amino-terminal gas-phase micro-sequencing (ABI 470A, Applied Biosystems, Foster City, CA; and PI 2090, Porton Instruments, Tarzana, CA), and the determined partial sequence (25 residues) was found to be identical to the sequence

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deduced from cDNA.¹ Additionally, a degraded HRG of 40,000 MW, which bound to DEAE-Sephacel and was eluted by a higher salt concentration than that of native HRG,¹⁵ also possessed an identical amino-terminal sequence, as determined by gas-phase micro-sequencing. The purified HRG preparations were not contaminated with bacterial lipopolysaccharide (LPS; <0.01 ng/ml), as determined by a chromogenic *Limulus amoebocyte* lysate microassay (Whittaker, Walkersville, MO). HRG preparations were extensively dialysed against RPMI-1640 medium prior to use in order to prevent possible interaction between HRG and amino acids, especially when [³⁵S]methionine was included in the metabolic labelling studies.

Macrophage FcγR-mediated phagocytosis

Photometric measurement of mouse inflammatory peritoneal macrophage FcγR-mediated binding and phagocytosis of antibody-conjugated sheep erythrocytes (EA) was performed as described previously.^{15,16} Briefly, inflammatory peritoneal macrophages were harvested from male C3HeB/FeJ mice (Jackson Laboratory, Bar Harbor, ME) with a 3-day prior intraperitoneal injection of 2 ml sterile paraffin oil. Macrophages (2×10^6 /ml), suspended in endotoxin-free RPMI-1640 supplemented with 200 mM L-glutamine, 25 mM HEPES and 50 µg/ml gentamycin, were seeded in 96-well flat-bottomed microtitre plates (Corning Glass Works, Corning, NY). Following adherence for 1 hr at 37°, the non-adherent cells were removed by washing with RPMI-1640. Macrophages were treated with native HRG, the degraded HRG, or medium alone for 24–48 hr, prior to exposure to EA for measuring the efficiency of FcγR-mediated binding and phagocytosis.^{15,16} Where indicated, macrophages were treated with HRG for 1–2 hr, followed by the removal of HRG. The culture was continued for an additional 24 hr without HRG prior to conducting phagocytic assays. To avoid interactions between HRG and serum proteins,^{2,9,10,15} macrophages were treated with purified HRG under serum-free conditions. More than 98% of macrophages remained viable after culturing in serum-free conditions for 48 hr, as determined by ⁵¹Cr-release assays.¹⁵ Quantification of macrophage FcγR-mediated phagocytosis of EA was measured at 405 nm for the absorbance of haemoglobin using an MR580 microtitre plate reader (Dynatech, Alexandria, VA). Percentages of phagocytic indices were calculated as described elsewhere.^{15,16}

Metabolic labelling

To assess the effect of HRG on cellular protein synthesis, human monocytic U937 (ATCC no. CRL1593; American Type Culture Collection, Rockville, MD), THP-1 (ATCC no. TIB202), and mouse peritoneal macrophages (from C3HeB/FeJ), suspended in methionine-free RPMI-1640 (Sigma) in the presence or absence of 10% fetal bovine serum (FBS), were treated with HRG for various indicated times at 37° in the presence of [³⁵S]methionine (ICN, Costa Mesa, CA). After incubation, the culture supernatants were harvested and the cells were lysed with 10 ml of a lysis buffer, 5 mM Tris-HCl, pH 7.4, containing 0.02% Na₃N, 10 mM EDTA, 5 mM iodoacetamide, 5 mM benzamidine, 0.15 M NaCl, 0.05% Nonidet-40 (NP-40) and 0.01% SDS. One hundred-microlitre aliquots of each cell lysate and supernatant was analysed twice by SDS-PAGE, and the dried gels were exposed to X-OMAT AR films (Kodak, Rochester, NY) overnight. The changes in the levels of intracellular or secreted proteins from HRG-treated and control cells

were determined by scanning the X-ray films using the Gilford Response spectrophotometer (Ciba Corning Diagnostics, Medfield, MA). The amount of protein synthesized from control cells was regarded as 100%. Alternatively, newly synthesized proteins from cell lysates or culture supernatants were precipitated by 10% trichloroacetic acid (TCA), followed by washing the pellets three times with H₂O and scintillation counting.

FcγRII expression: cellular ELISA and immunoprecipitation

To determine the effect of HRG on the expression of membrane type II Fcγ receptor (FcγRII; CD32),¹⁷ THP-1 cells (2×10^6 /ml) were seeded onto 96-well microtitre plates for 24 hr, followed by treating with purified human HRG (40 ng/ml–4 µg/ml) for 16–20 hr in the presence or absence of 10% FBS. After incubation, the cells were fixed with 0.5% formaldehyde and the cell-surface FcγRII levels were measured by enzyme-linked immunosorbent assay (ELISA). The entire procedure was performed according to the instructions of the Pierce ELISA kit (Pierce, Rockford, IL). Both monoclonal antibodies against CD32 (AMAC, Westbrook, IL) and anti-mouse IgG antibodies conjugated with alkaline phosphatase (Pierce) were utilized. The effect of HRG on the proliferation of THP-1 cells after 24 hr treatment was determined by staining the viable cells with crystal violet.¹⁸

The effect of HRG on FcγRII expression was also determined by metabolic labelling. THP-1 cells (2×10^6 /ml) were treated with purified human HRG (40 ng/ml–4 µg/ml) for 18 hr in the presence or absence of 10% FBS. The cells were then washed with RPMI-1640 twice to remove HRG and labelled with [³⁵S]methionine (100 µCi/10⁶ cells) for 2 hr at 37° in methionine-serum-free RPMI-1640 (Sigma, St Louis, MO). The cells were then washed once with cold RPMI-1640 and lysed with 10 ml of the lysis buffer as indicated above. The cell lysates were precleared for 1 hr at 4° by adding 1 mM phenylmethylsulphonyl fluoride (PMSF, Sigma), 10 µg/ml aprotinin (Sigma) and 30 µl of protein A agarose beads (Pierce). Following centrifugation at 1500 g at 4° for 15 min to remove the beads and cellular debris, FcγRII was immunoprecipitated by adding 2 µl of anti-CD32 monoclonal antibodies (1 µg/µl; AMAC Inc., Westbrook, ME) and 10 µl of protein A agarose beads to the precleared cell lysates for rotating end-over-end at 4° for 4 hr. The protein A-bound materials were washed three times with 5 ml of 10 mM Tris-HCl, pH 8.0, containing 1 mM EDTA, 10 mM Na₃N, 0.25% NP-40, 100 µM PMSF and 0.2 M NaCl, and released by heating at 100° for 30 seconds in 100 µl of 0.06 M Tris-HCl, pH 6.8, 0.1% glycerol, 2.0% SDS and 5% β-mercaptoethanol. SDS-PAGE was performed to analyse FcγRII and the dried gel was exposed to an X-OMAT AR film at –70° for 1 week. HRG-mediated changes in FcγRII expression relative to control cells were analysed by scanning the X-ray film.

RESULTS

Regulation of FcγR-mediated phagocytosis

Previously we reported that freshly harvested mouse peritoneal inflammatory macrophages, when pretreated with HRG under serum-free conditions for 1–2 hr at 37°, increased their binding and phagocytic efficiency of EA cells by 10–50%.¹⁵ In contrast, a long-term pretreatment of macrophages with HRG for 16–24 hr resulted in down-regulation of FcγR-mediated EA binding/

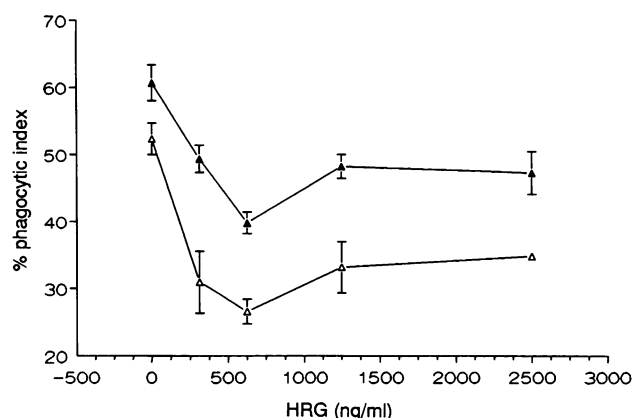


Figure 1. Effect of mouse HRG on macrophage phagocytosis. Freshly harvested mouse peritoneal inflammatory macrophages were pretreated with mouse HRG (312.5 ng/ml–2.5 μ g/ml) for 1 hr (Δ) or 2 hr ($\blacktriangle$) under serum-free conditions, followed by washing the monolayers with RPMI-1640 and continuing the cell culture without HRG for an additional 24 hr. Macrophage Fc γ R-mediated phagocytosis was performed by adding 100- μ l aliquots of EA cells [sheep erythrocytes conjugated with polyclonal IgG against sheep erythrocytes (1/5000 dilution)] to the macrophages, followed by 1 hr incubation at 37°. The percentage of the phagocytic index was calculated as previously described.¹⁵ Each data point represents mean percentage phagocytic index $\pm$ SD from quadruplicate experiments.

phagocytosis by 40–90%. As further determined in this study, a short-term exposure of macrophages to HRG was sufficient to reduce their phagocytic function. For example, pretreatment of macrophages with HRG for 1–2 hr, followed by culturing the cells without HRG for an additional 24 hr, resulted in reduction of Fc γ R-mediated EA phagocytosis (Fig. 1). At low concentrations (less than 600 ng/ml), HRG decreased the phagocytic functions in a linearly dose-dependent manner. However, by increasing the concentrations up to 2.5 μ g/ml, the inhibitory effect of HRG was partially reversed. Percentages of EA binding to the HRG-pretreated macrophages at 4° for 1 hr were also reduced (data not shown). HRG was not cytotoxic to macrophages and the reduction of macrophage phagocytic function was not due to cell death, as determined by ⁵¹Cr-release assay.

The reduction of macrophage phagocytic function by HRG was found to be reversible (Fig. 2). For example, macrophage Fc γ R-mediated phagocytosis was reduced by HRG after 24 hr treatment, and the reduction was partially reversed when the treatment was prolonged up to 48 hr. Moreover, by pretreating macrophages with HRG for 24 hr, followed by removing HRG and continuing the culture for another 24 hr, the Fc γ R-phagocytic function was also partially restored, compared to the macrophage function after 24 hr treatment (Fig. 2).

The freshly harvested peritoneal inflammatory macrophages were precultured *in vitro* at 37° for 24–48 hr prior to testing their responsiveness to HRG. It was determined that the freshly harvested macrophages after 24 hr in culture could still respond to up- or down-regulation of Fc γ R-mediated phagocytosis by HRG after 1 and 24 hr treatment, respectively (Fig. 3). The percentages of increases in Fc γ R-dependent phagocytosis were comparable to the results obtained when the freshly harvested peritoneal macrophages were directly treated with HRG for 1–2 hr, followed by performing phagocytic assays.¹⁵ Similar results were obtained by testing the function of freshly harvested

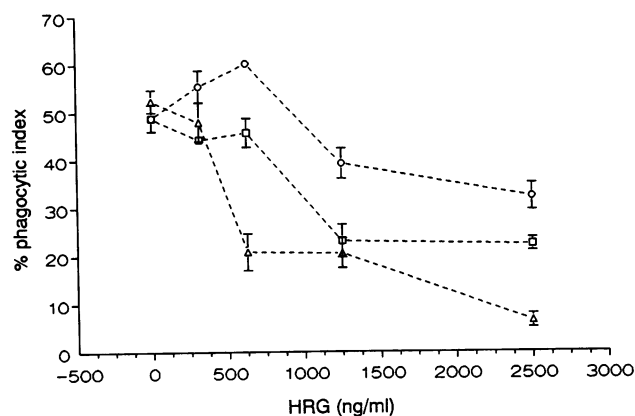


Figure 2. Reversible effect of long-term pretreatment of macrophages with HRG on phagocytic functions. Mouse peritoneal inflammatory macrophages were treated with purified mouse HRG for 24 hr (Δ) or 48 hr ($\square$), followed by testing their phagocytic efficiency. Additionally, macrophages were treated with HRG for 24 hr, followed by removing HRG and continuing the cell culture for an additional 24 hr prior to phagocytic assays ($\circ$). Each data point represents mean percentage phagocytic index $\pm$ SD from quadruplicate experiments.

peritoneal macrophages cultured *in vitro* for up to 48 hr (data not shown). Compared with mouse plasma HRG, human plasma HRG was 10–20% less effective in reducing murine macrophage Fc γ R-mediated phagocytosis. Such reduction was also reversible (data not shown).

Localization of HRG functional domain

The N-terminal sequence of the degraded HRG (40,000 MW) was identical to that of the native HRG, as determined by N-terminal gas-phase micro-sequencing. The degraded HRG from mouse plasma was shown to reduce Fc γ R-mediated EA phagocytosis after treating macrophages for 16 hr in a dose-related manner (Fig. 4a). On a molar basis, the degraded HRG was approximately four-fold less effective in reducing macrophage phagocytosis than that of native HRG. Binding of EA cells to

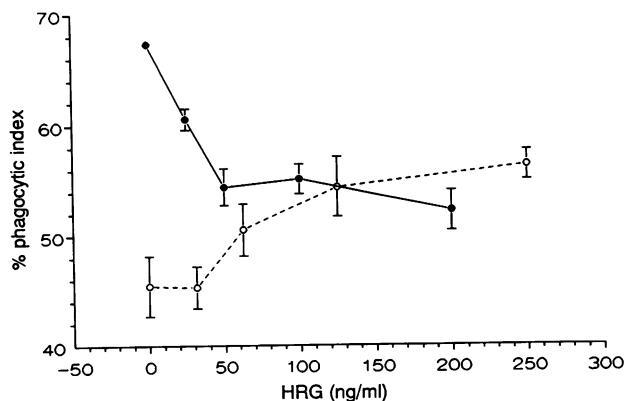


Figure 3. Effect of HRG on precultured macrophages. Freshly harvested mouse peritoneal macrophages were cultured 24 hr prior to incubation with mouse HRG for 1 hr ($\circ$) or 24 hr ($\bullet$). Phagocytosis of EA cells (sheep erythrocytes conjugated with 1/4000 diluted IgG) by macrophages was then performed in quadruplicates by the standard procedure.

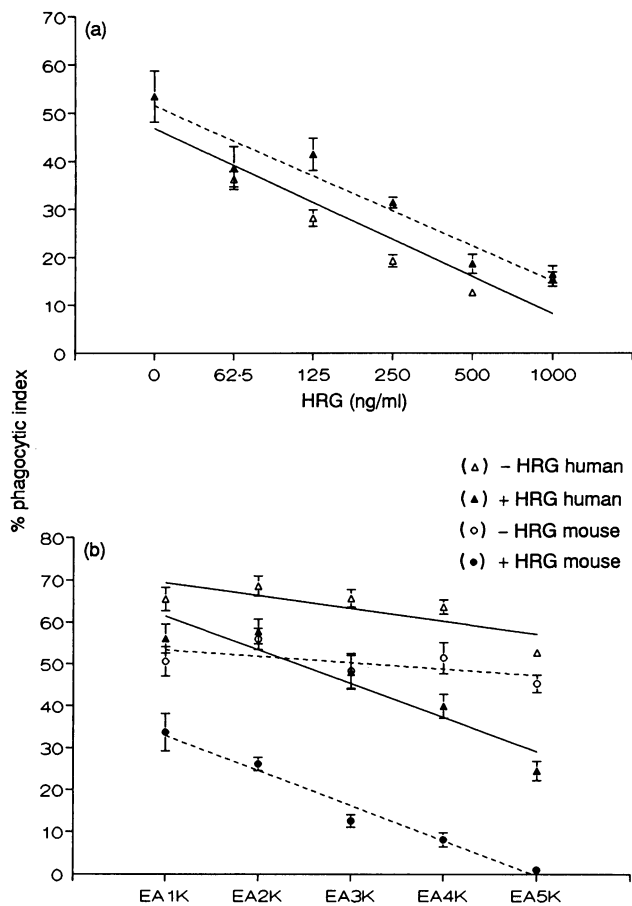


Figure 4. Effect of degraded HRG on macrophage phagocytosis. (a) Freshly harvested mouse peritoneal inflammatory macrophages were treated with mouse native HRG (Δ) or degraded HRG ($\blacktriangle$) for 16 hr, followed by testing their phagocytic efficiency for EA (sheep erythrocytes conjugated with 1/4000 diluted IgG). (b) Additionally, macrophages were treated with degraded mouse HRG (100 ng/ml) or degraded human HRG (100 ng/ml) for 20 hr prior to conducting phagocytic assays in quadruplicates. Sheep erythrocytes were coated with various dilutions of IgG: EA1K, 1/1000 diluted IgG; EA2K, 1/2000 diluted IgG; EA3K, 1/3000 diluted IgG; EA4K, 1/4000 diluted IgG; EA5K, 1/5000 diluted IgG.

the degraded HRG-pretreated macrophages at 4° was also reduced (data not shown). Similarly, degraded HRG from human plasma was capable of reducing mouse macrophage FcγR-mediated phagocytosis, but with less efficiency compared to mouse-degraded HRG (Fig. 4b).

Modulation of FcγRII expression by HRG

By cellular ELISA, it was determined that HRG decreased the expression of FcγRII on THP-1 cells in a dose-related manner after 18 hr treatment in the absence of serum (Fig. 5a). Similar results were obtained when THP-1 cells were treated with HRG for 18 hr and simultaneously labelled with [³⁵S]methionine in the absence of serum, followed by immunoprecipitation (data not shown).

However, when THP-1 cells were treated with HRG for 18 hr under serum-free conditions, followed by removing HRG and labelling with [³⁵S]methionine for 2 hr without serum, the

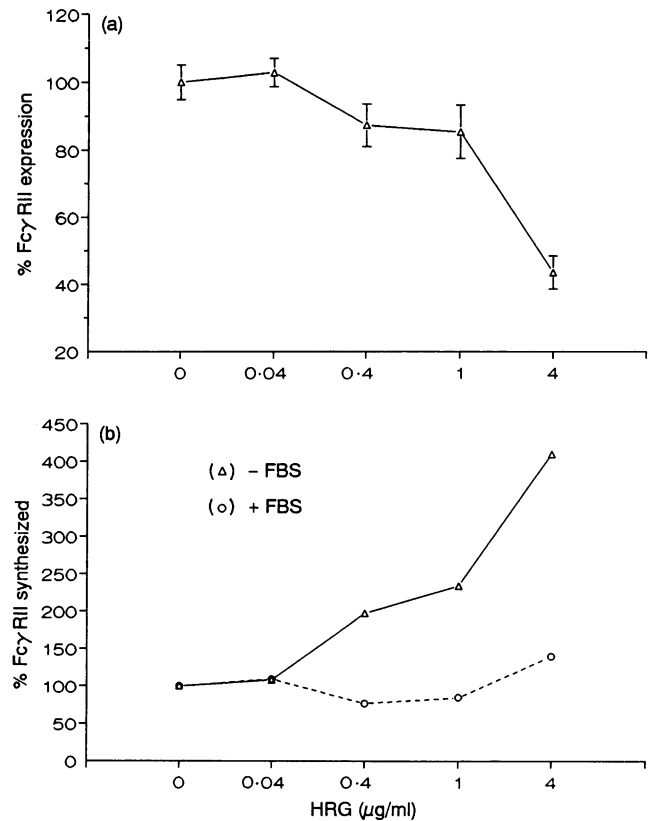


Figure 5. Effect of HRG on FcγRII expression. (a) THP-1 cells (1×10^6 /ml) were pretreated with human HRG for 18 hr under serum-free conditions, followed by fixing the cells with 0.5% formaldehyde and detecting cell-surface FcγRII by ELISA using monoclonal anti-CD32 antibodies. Each data point represents an average ($\pm$ SD) of 12 duplicate experiments. (b) THP-1 cells were treated with human HRG for 18 hr in the absence or presence of 10% FBS, followed by washing and labelling for 2 hr with [³⁵S]methionine. Membrane FcγRII was solubilized, immunoprecipitated and quantified (see the Materials and Methods). The FcγRII level in each control cell group was regarded as 100%. Only one set of data from two experimental runs is presented.

HRG-treated cells had an enhanced capacity of synthesizing membrane FcγRII compared to control cells (Fig. 5b). In the presence of serum during pretreatment with HRG for 18 hr, the capacity of THP-1 cells in synthesizing membrane FcγRII was reduced (Fig. 5b).

In addition to the membrane FcγRII of 40,000 MW, there was an additional FcγRII species of 45,000 MW found in the immunoprecipitate. The 45,000 MW FcγRII probably corresponds to the intracellular precursor. The amount of intracellular pool of the 45,000 MW FcγRII was approximately half of the membrane form, and HRG also increased the expression of the FcγRII precursor to a similar extent as that of the membrane form (data not shown).

HRG-mediated modulation of FcγRII expression was not due to its regulation of cellular proliferation. HRG (40 ng/ml–4 μg/ml) could not enhance or reduce THP-1 cell growth after 24 hr treatment ($P > 0.1$; experiments versus controls).

Effect of HRG on protein synthesis and secretion

The capacity of HRG to regulate cellular protein synthesis and secretion was examined, since one or both processes could

Table 1. Effect of HRG on U937 cell protein secretion*

		[³⁵ S]methionine incorporation (c.p.m.)	% inhibition of protein secretion
Control		54,678	
HRG	0.425 µg/ml	74,162	-35.6
	0.850 µg/ml	48,128	+12.0
	1.700 µg/ml	33,504	+38.7

* U937 cells (8×10^5 /ml in methionine-free RPMI-1640) were treated with various quantities of HRG for 24 hr in the presence of 30 µCi [³⁵S]methionine. After incubation, aliquots of culture supernatants were recovered, precipitated with 10% TCA, and counted in a scintillation counter. + indicates reduction in protein secretion, whereas - represents enhancement of protein secretion. More than 98% of both control and HRG-treated cells remained viable after 24 hr in culture. Each value represents an average of two experimental results.

account in whole or in part for the modulation of macrophage FcγR expression and phagocytic function. As demonstrated in the Table 1, HRG either enhanced or inhibited protein secretion in monocytic U937 cells, depending upon the concentration of HRG used. The results were further confirmed by SDS-PAGE

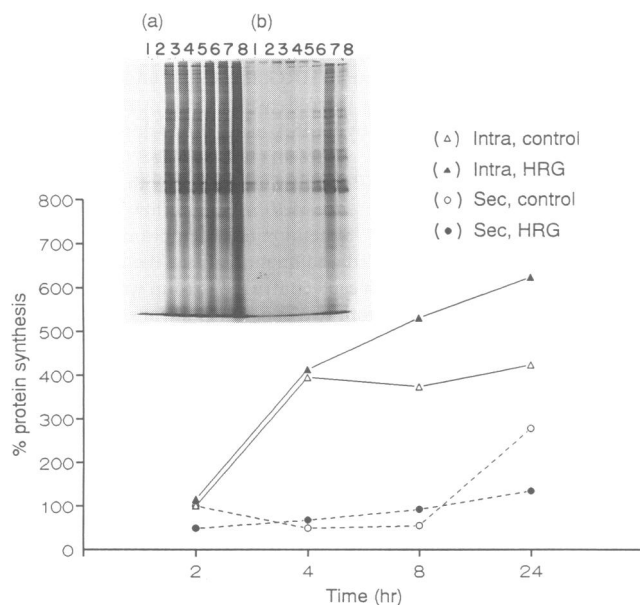


Figure 6. Effect of human HRG on protein synthesis and secretion by monocytic U937 cells. U937 cells (7×10^5 /ml) were treated with 1 µg/ml human HRG for various times at 37° in the presence of [³⁵S]methionine. After treatment, 100-µl aliquots of the cell lysates (intracellular) and culture supernatants (secreted) were subjected to SDS-PAGE and autoradiography. The X-ray film was then scanned in an automatic spectrophotometer. The control cells at 2 hr labelling were regarded as having 100% protein synthesis, either intracellularly or extracellularly (secreted). The level of protein synthesis measured at other indicated times was compared to the 2-hr time-point of the control cells. Intra, intracellular; Sec, secreted. In SDS-PAGE: (a) intracellular; (b) secreted. Controls: 1, 3, 5, 7. HRG-treated: 2, 4, 6, 8. 2 hr treatment: 1, 2. 4 hr: 3, 4. 8 hr: 5, 6. 24 hr: 7, 8. Only one set of data from two experimental runs is presented.

and autoradiography (data not shown). More than 98% of both control and HRG-treated monocytic U937 cells remained viable after 24 hr in culture. To further evaluate the effect of HRG on protein synthesis, time-course metabolic labelling studies were carried out. Under serum-free conditions, HRG enhanced cellular protein synthesis of U937 cells in a time-dependent manner over 8–24 hr. It resulted in an accumulation of intracellular proteins but decreased protein secretion after 24 hr of culturing, compared to untreated control cells (Fig. 6). As an appropriate control, U937 cells were treated with albumin in the metabolic labelling studies. Albumin had no apparent effect on altering the levels of protein synthesis and/or secretion in U937 cells (data not shown). In addition, when U937 cells were pretreated with HRG for 24 hr, followed by metabolic labelling for 2 hr with [³⁵S]methionine, the HRG-pretreated cells had an enhanced capability of synthesizing proteins (data not shown). In contrast, when U937 cells were treated with HRG in the presence of 10% FBS, the exogenous HRG suppressed total protein synthesis with time, which resulted in approximately 25% reduction in protein synthesis for up to 8–24 hr of treatment. Similar results were obtained using human monocytic THP-1 cells and mouse macrophages (data not shown).

DISCUSSION

The intriguing finding in this study is that HRG signalled an increase in the expression of FcγR shortly after treating macrophages with HRG for 1–2 hr, followed by a significant reduction in FcγR levels when the HRG-pretreated cells were continued in culture for an additional 16–48 hr without HRG. The molecular mechanisms by which HRG regulates macrophage functions are unknown. However, it is reasonable to assume that HRG regulates macrophages via a receptor-dependent signal transduction mechanism. Specific saturation binding of radio-iodinated HRG to macrophages supports this hypothesis (N.-S. Chang, manuscript in preparation). Freshly harvested peritoneal macrophages were still responsive to HRG stimulation even after 24–48 hr in culture. The results imply that macrophages in a more differentiated state, such as an increased expression of membrane FcγR after 24 hr in culture,¹⁵ still possess receptor sites for HRG.

Receptor-dependent cellular response has been implicated in the regulation of T-cell function by HRG.¹⁹ Two classes of HRG-binding sites on T cells have been found; however, only one binding protein of 56,000 MW is identified on T-cell membrane.¹⁹ The molecular domain of HRG responsible for binding to T cells and macrophages has not been characterized. Our studies demonstrated that the degraded HRG of 40,000 MW, which possessed an identical N-terminal sequence as the native HRG, was capable of regulating macrophage phagocytic function. This observation indicates that the functional domain of HRG is located at the N-terminal half.

Functionally, this C-terminally deleted HRG is approximately four-fold less efficient in restricting macrophage phagocytosis compared to native HRG, suggesting that a certain portion of C-terminal domain is also essential in the regulation of cellular functions. The carbohydrate content of HRG is only 14%.^{1,2} By stripping all the carbohydrates off the native HRG, the molecular size becomes 66,000–70,000 MW, which is still larger than the degraded 40,000 MW HRG. Thus, it is unlikely that the degraded HRG is a glycosylated variant of native HRG.

Conceivably, the degraded HRG has a major carboxylic portion (approximately 40,000 MW) deleted from the native protein, probably including the far-end C-terminal histidine-rich region (no. 330–389 within a total 507 amino acids of HRG).

HRG was found to be an effective regulator for protein synthesis and/or secretion in macrophages and monocytic cell lines. For example, HRG increased total cellular proteins in U937 cells after 1–8 hr treatment under serum-free conditions. Accumulation of intracellular proteins was observed upon prolonged treatment with HRG for 8–24 hr, whereas there was a reduced level of proteins found in the secretion. Reduction of protein secretion by HRG implies that (1) HRG alters the functions of protein secretory machinery, probably including both endoplasmic reticulum and the trans-Golgi network; (2) HRG enhances the turnover rates of intracellular proteins; or (3) the accumulated intracellular proteins block the secretory paths.

Based on the premise that HRG alters cellular metabolism, the reduction of macrophage FcγR expression and phagocytic function may directly correlate with the regulation of protein synthesis and/or secretion by HRG. For example, increases in membrane FcγR levels during an initial 1–4 hr treatment of mouse macrophages with HRG were positively correlated with the enhancement of protein synthesis. However, during a long-term treatment with HRG, mouse macrophages appeared to acquire an increased capacity for synthesizing cellular proteins, but a reduced ability in FcγR expression and phagocytic function. Similar results were obtained using less differentiated human monocytic cells regarding protein synthesis/secretion and FcγRII expression. Thus, the reduction of FcγR expression suggests that (1) there was a reduced rate of FcγR translocation from endoplasmic reticulum to cell membrane; or (2) there was an enhanced turnover rate for FcγR proteins.

In some cases, changes in macrophage functions were not positively correlated with the stimulation of HRG in a dose-dependent manner. For instance, when murine macrophages were treated with HRG for 1–2 hr, followed by 24 hr of culturing without HRG, the phagocytic function of the cells was not decreased linearly by HRG in a dose-dependent manner (Fig. 1). A possible explanation is that when macrophages were exposed to a relatively high concentration of HRG, the cells probably had acquired a higher capacity of protein synthesis than when exposed to a lower concentration of HRG. However, additional functional mechanisms may also be involved, such that the expression and function of FcγR were manifested in diverse ways.

Exogenous HRG, when combined with 10% heat-inactivated FBS bovine serum, down-regulated protein synthesis in U937 cells. Furthermore, in the presence of serum the enhanced capacity of THP-1 cells in synthesizing FcγRII was decreased. A possible explanation is that the exogenous HRG, together with the residual functional endogenous HRG after heat-treatment of serum, may add up to an effective concentration that restricts cellular functions. Furthermore, HRG is capable of forming complexes with several serum proteins such as plasminogen,² thrombospondin⁹ and S-protein (vitronectin).¹⁰ Exogenous HRG when added to serum would shift the chemical balance towards the formation of these complexes. The functional properties of HRG may be neutralized by these binding proteins, as evidenced by decreases in its effect on up-regulation of FcγRII biosynthesis in THP-1 cells. Alternatively, complex

formation allows exposure of new epitopes in HRG, and the conformationally altered HRG would assume a new function in the regulation of cellular protein synthesis and gene expression.

Although an apparent overall protein synthesis may be up-regulated or down-regulated by HRG, depending upon the presence or absence of serum, it may not necessarily indicate that biosynthesis of all the protein species is affected to the same extent by HRG. For example, in our previous study, HRG differentially regulated the expression and function of FcγR subclasses on mouse macrophages.¹⁵ HRG reduced FcγRII (Fcγ2b/γ1R)-dependent phagocytosis more effectively than that of FcγRI (Fcγ2aR).¹⁵

Liver appears to be the major site for HRG biosynthesis. Whether monocytic cells synthesize HRG remains to be established. However, it was found that antibodies against HRG could bind to the THP-1 cell surface, as determined by cellular ELISA; there were two-fold more antibodies binding to LPS/IFN-γ-treated THP-1 cells than to control cells (N.-S. Chang, unpublished data). The functional significance of these observations is unknown. However, the findings indicate that THP-1 cells could synthesize HRG or HRG-like molecules. These molecules may be important in modulating macrophage functions in a localized inflammatory microenvironment.

Since HRG is present at a substantial concentration in plasma (100–150 μg/ml), it probably plays a significant role in maintaining normal physiological equilibrium. An elevation of plasma HRG is believed to be associated with thrombophilia.²⁰ Moreover, HRG has been shown to be a negative acute phase reactant.²¹ That is, reduction of plasma HRG occurs during the acute phase. Such a reduction may be necessary for facilitating the acute inflammatory response. In our previous report,¹⁰ a balanced concentration of plasma HRG was found to be critical in optimizing complement functional efficiency. Morgan reported that serum HRG levels are decreased in patients with acquired immune deficiency syndrome or those immunosuppressed when subjected to steroid therapy.²² Conceivably, a steady level of plasma HRG is essential in controlling blood T lymphocyte and monocyte functions.

In addition, to macrophages, HRG has been shown to suppress T-cell proliferation, inhibit expression of interleukin-2 receptor on activated T cells, reduce T-cell IFN-γ release and alter T-cell dependent inhibition of erythropoiesis.¹⁴ Most recently, HRG was found to stimulate MOLT-3 (a T-cell line) and a subpopulation of peripheral blood T cells to adhere and extend processes (R. J. Lamb and W. T. Morgan, submitted for publication). The molecular action by which HRG regulates T-cell function is unknown. However, based on the observations in this study, it is suggested that HRG could also regulate cellular protein synthesis and/or secretion in T cells.

Overall, in this study we have further confirmed the regulatory function of HRG on macrophages and localized the functional domain to the N-terminal half of HRG. Regulation of protein synthesis and/or secretion in macrophages or monocytic cell lines by HRG accounts in part for modifying biological functions in these cells. The signal transduction mechanisms in response to HRG is currently being investigated.

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